

Distributions



Binomial distribution

Excel function **=BINOMDIST (x, n, π , cumulative)**

computes Binomial probabilities. For example, use

=BINOMDIST (7, 12, 0.6, 0) to find the probability of exactly $x=7$ successes in $n=12$ Bernoulli trials, where the probability of a successful trial is $\pi = 0.6$, and use

=BINOMDIST (7, 12, 0.6, 1) to find the probability of no more than $x=7$ successes (that's cumulative).

Poisson distribution

Excel function **=POISSON (x, μ , cumulative)**

computes Poisson probabilities. For example, use

=POISSON (7, 12, 0) to find the probability of exactly $x=7$ "rare" events when the average frequency of events is $\mu=12$, and use

= POISSON (7, 12, 1) to find the probability of no more than $x=7$ "rare" events (that's cumulative).

Normal distribution

Excel function **=NORMDIST (x, μ , σ , 1)**

computes Normal probabilities. For example, use

=NORMDIST (25, 30, 4, 1) to find the probability of $X \leq 25$ for a Normal random variable with mean $\mu=30$ and standard deviation $\sigma=4$.

(Command =NORMDIST(x, μ , σ , 0) computes the Normal density. In this course, you won't need it)

Also, Excel function **=NORMINV (p, μ , σ)** solves the inverse problem, finding the value of x that corresponds to the given probability p.

For example, use

=NORMINV (0.25, 30, 4, 1) to find the value x so that a Normal random variable with mean $\mu=30$ and standard deviation $\sigma=4$ does not exceed x with probability 0.25.